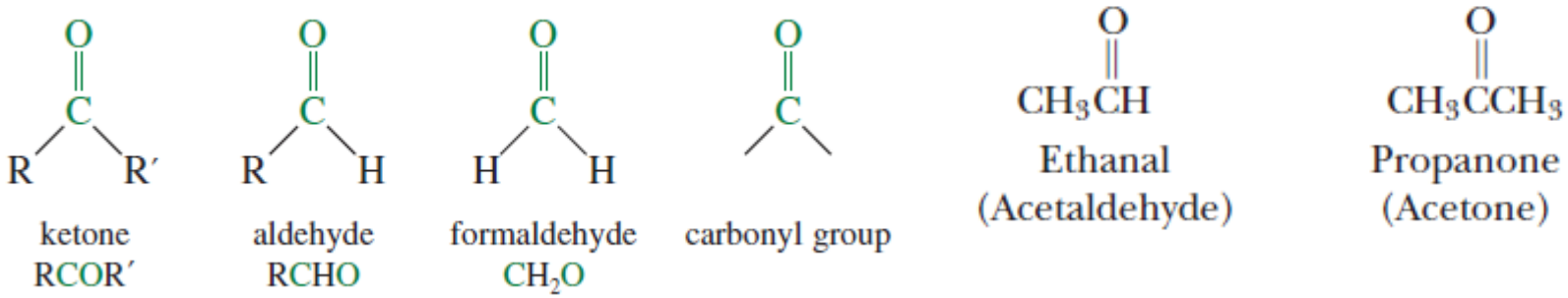
Aldehydes and Ketones

Dr Fatema Jagot

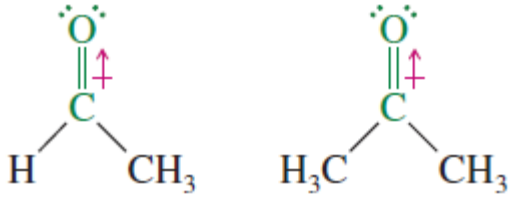
Aldehydes and ketones

- The simplest carbonyl compounds are the aldehydes and ketones.
- The functional group of an aldehyde is a carbonyl group bonded to a hydrogen atom.
- In methanal (formaldehyde), the simplest aldehyde, the carbonyl is bonded to two hydrogen atoms. In other aldehydes besides, the carbonyl group is bonded to a hydrogen atom and a carbon atom.
- The functional group of a ketone is a carbonyl group bonded to two carbon atoms.



The carbonyl group

- The carbonyl carbon atom is sp^2 hybridised and is bonded to three other atoms through coplanar sigma bonds about 120° apart (trigonal planar geometry).
- The unhybridised *p* orbital overlaps with the *p* orbital of oxygen to form a *pi* bond.
- The double bond between the carbon and oxygen is similar to an alkene C=C double bond, except that the C=O bond is shorter, stronger, and more polarised.



The polarity of the C=O bond contributes to the reactivity of aldehydes and ketones. The negatively polarised oxygen acts as a **nucleophile**, and the positively polarised carbon acts as an **electrophile**.

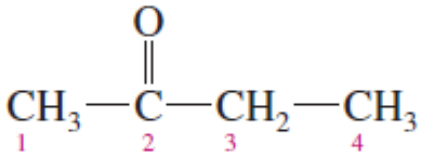
Common Classes of Carbonyl Compounds

Class	General Formula	Class	General Formula
ketones	$R-\overset{\overset{\large O}{\|}}{C}-R'$	aldehydes	$R-\overset{\overset{\large O}{\|}}{C}-H$
carboxylic acids	$R-\overset{\overset{\large O}{\|}}{C}-OH$	acid chlorides	$R-\overset{\overset{\large O}{\|}}{C}-Cl$
esters	$R-\overset{\overset{\large O}{\|}}{C}-O-R'$	amides	$R-\overset{\overset{\large O}{\|}}{C}-NH_2$

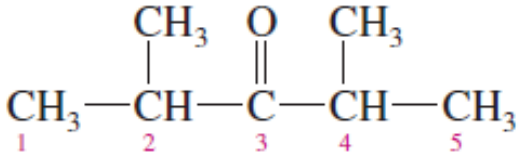
Nomenclature of aldehydes and ketones

- Systematic names for ketones are obtained by replacing the final ‘e’ of the parent alkane with ‘one’. So alkane name becomes alkanone.
- In open chain ketones, the parent chain is the longest chain containing the carbonyl group.
- The chain is numbered from the end that gives the lowest number for the carbonyl group.
- In cyclic ketones, the carbon of the carbonyl group is assigned the number 1.

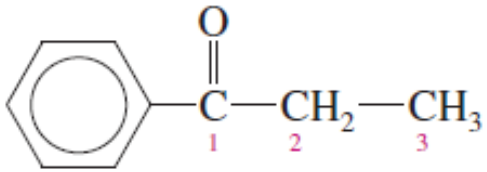
Priority of Functional Groups in Naming Organic Compounds
(highest) acids
esters
aldehydes
ketones
alcohols
amines
alkenes, alkynes
alkanes
ethers
(lowest) halides



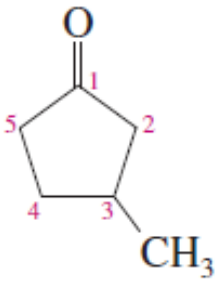
2-butanone
butan-2-one



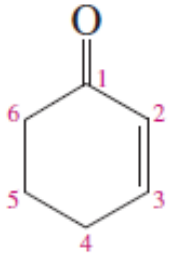
2,4-dimethyl-3-pentanone
2,4-dimethylpentan-3-one



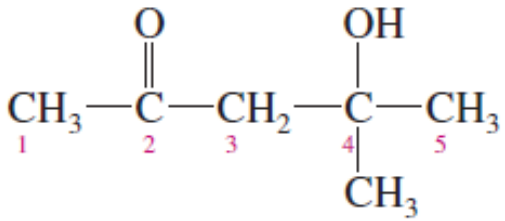
1-phenyl-1-propanone
1-phenylpropan-1-one



3-methylcyclopentanone

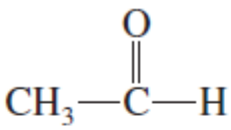


2-cyclohexenone
cyclohex-2-en-1-one

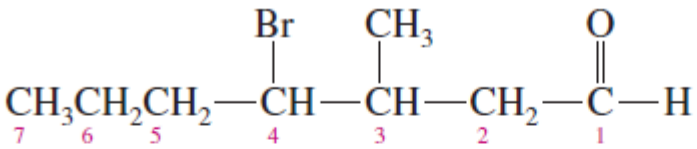


4-hydroxy-4-methyl-2-pentanone
4-hydroxy-4-methylpentan-2-one

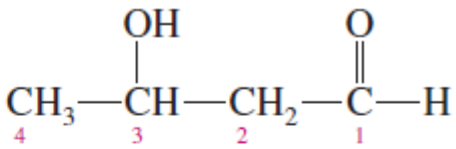
- Systematic names for aldehydes are derived by replacing the final ‘e’ of the alkane with ‘al’. So the alkane name becomes alkanal.
- An aldehyde at the end of the chain is carbon number 1.
- If the aldehyde is a substituent of a large unit (usually a ring), it is given the suffix ‘*carbaldehyde*’.



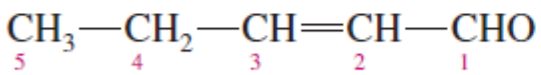
ethanal



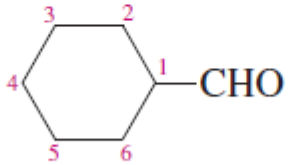
4-bromo-3-methylheptanal



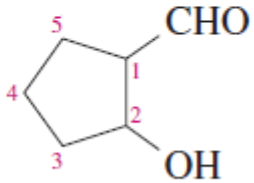
3-hydroxybutanal



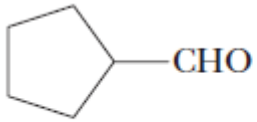
2-pentenal
pent-2-enal



cyclohexanecarbaldehyde



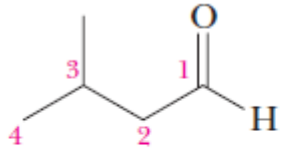
2-hydroxycyclopentane-1-carbaldehyde



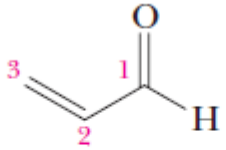
Cyclopentanecarbaldehyde

trans-4-Hydroxycyclohexanecarbaldehyde

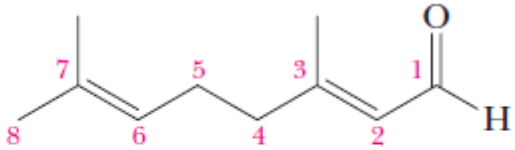
- For **unsaturated aldehydes**, the presence of the double bond is indicated by the infix –en-.
- The location of the suffix determines the numbering pattern.



3-Methylbutanal

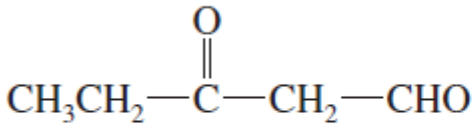


2-Propenal
(Acrolein)

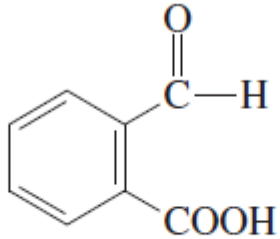


(2*E*)-3,7-Dimethyl-2,6-octadienal
(Geranial)

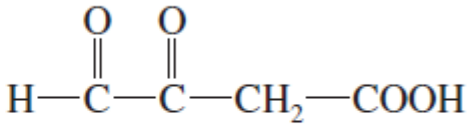
- A ketone or aldehyde group can also be named as a substituent on a molecule with a higher priority functional group as its root.
- A ketone or aldehyde carbonyl group is named by the prefix *oxo*- if it is included in the main parent chain.
- If the aldehyde is not included in the parent chain, it is given the prefix *formyl*-



3-oxopentanal

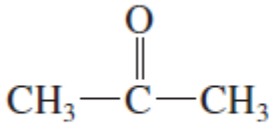


2-formylbenzoic acid

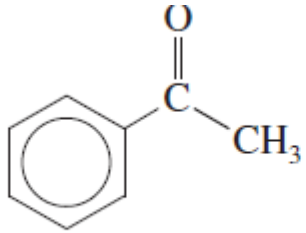


3,4-dioxobutanoic acid

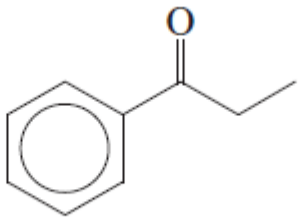
Some common names



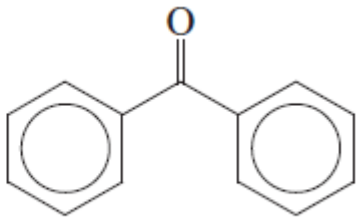
acetone



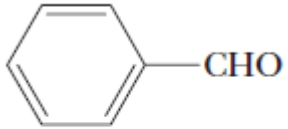
acetophenone



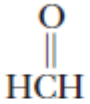
propiophenone



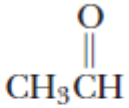
benzophenone



Benzaldehyde

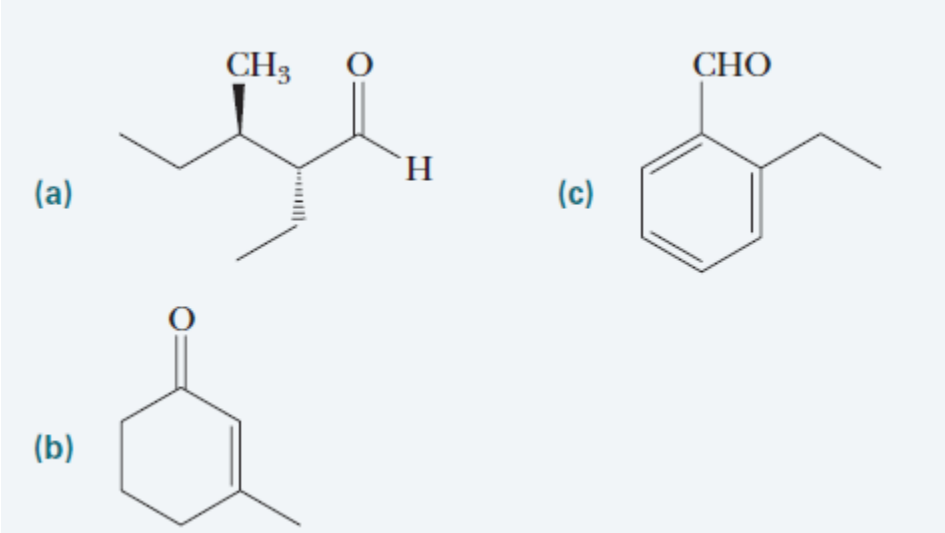


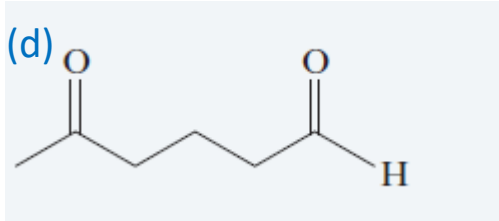
Formaldehyde

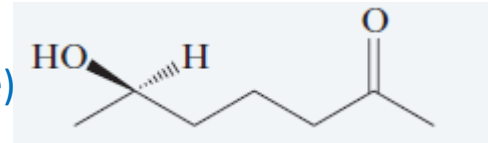


Acetaldehyde

- Practice exercise: Write the IUPAC name for each of the following







a) 2-ethyl-3-methylpentanal

b) 3-methyl-2-cyclohexenone

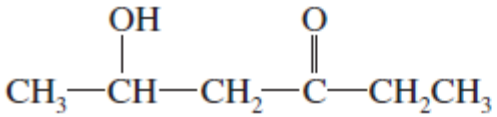
c) This molecule is derived from benzaldehyde. Its name is 2-ethylbenzaldehyde

d) 5-oxohexanal

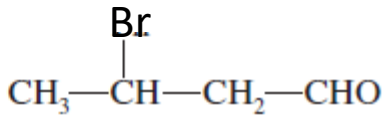
e) 6-hydroxy-2-heptanone.

- Practice exercise 2: Write the IUPAC name for each of the following:

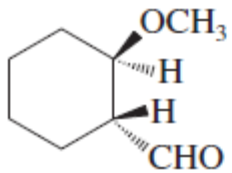
(a)



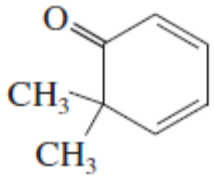
(b)



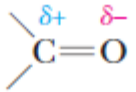
(c)



(d)



Physical properties of aldehydes and ketones



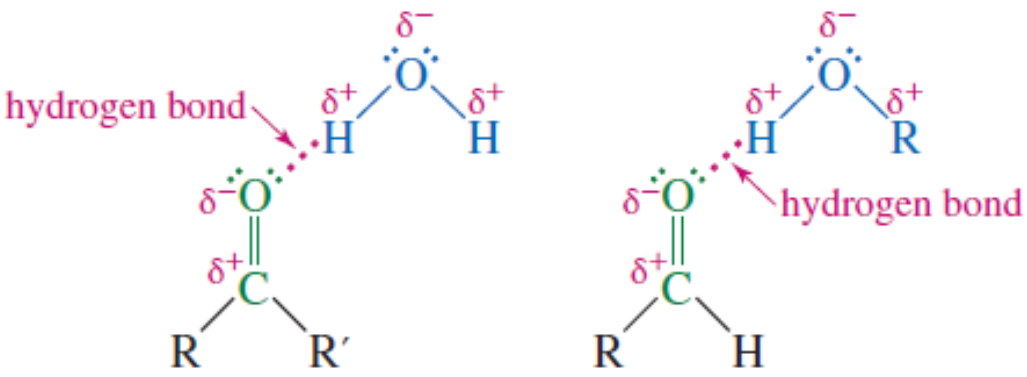
Polarity of a carbonyl group

- The carbon-oxygen bond is polar.
- Dipole-dipole interactions exist between ketone molecules and between aldehyde molecules.
- Therefore their boiling points are higher than those of hydrocarbons and ethers of comparable molecular weight.
- Aldehydes and ketones do not have any O—H or N—H bonds, and therefore they can not hydrogen bond with each other.

Their boiling points are lower than those of alcohols and carboxylic acids of comparable molecular weight due to the inability of aldehydes and ketones to hydrogen bond with each other. Alcohols and carboxylic acid molecules associate with each other through hydrogen bonding.

Boiling Points of Six Compounds of Comparable Molecular Weight			
Name	Structural Formula	Molecular Weight	Boiling Point (°C)
Diethyl ether	$CH_3CH_2OCH_2CH_3$	74	34
Pentane	$CH_3CH_2CH_2CH_2CH_3$	72	36
Butanal	$CH_3CH_2CH_2CHO$	72	76
2-Butanone	$CH_3CH_2COCH_3$	72	80
1-Butanol	$CH_3CH_2CH_2CH_2OH$	74	117
Propanoic acid	CH_3CH_2COOH	72	141

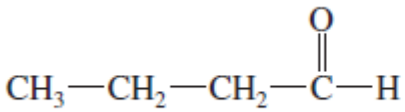
- Although aldehydes and ketones can not hydrogen bond with each other, they have lone pairs of electrons on the oxygen and can act as hydrogen bond acceptors with other compounds having O—H and N—H bonds.
- Because the carbonyl group of aldehydes and ketones can interact with water through hydrogen bonding, these compounds of low-molecular weight are soluble in water, and their solubility is greater than for non—polar molecules of comparable molecular weight.



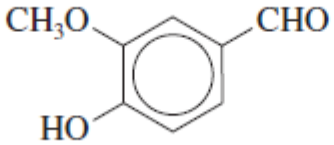
Uses of aldehydes and ketones

- Because of their ability to act as hydrogen bond acceptors, they make good solvents for polar hydroxylic substances such as alcohols.
- Formaldehyde and acetaldehyde are the most common aldehydes.
- Formaldehye in solution is used to preserve biological specimen.
- It is also used to make polymeric products, urea-formaldehyde glues, and resins.
- Acetaldehyde is used in the manufacture of acetic acid, polymers, and drugs.

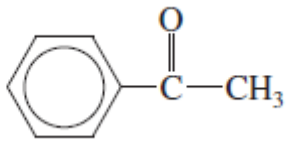
- Acetone is the most important commercial ketone, used as a solvent of low toxicity.
- Many other ketones and aldehydes are used as flavourings and additives to food, drugs, and other products. For example, benzaldehyde is the primary component in almond extract.



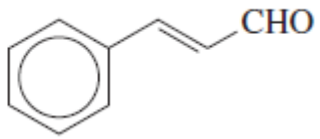
butyraldehyde



vanillin



acetophenone



trans-cinnamaldehyde

Odor:	buttery	vanilla	pistachio	cinnamon
Uses:	margarine, foods	foods, perfumes	ice cream	candy, foods, drugs